

Siddhartha Kumar

Software Engineer • Full-Stack Development • AI/ML

✉ devsiddhartha31@gmail.com 📞 +91 7260893320 📍 Bihar, India
🐙 github.com/devsiddhartha31 🔗 [linkedin.com/in/devsiddhartha31](https://www.linkedin.com/in/devsiddhartha31)

Profile

Software Engineer experienced in designing, building, and deploying production-grade full-stack applications, Android solutions, cloud infrastructure, and AI systems. Delivered scalable software for international digital campaigns and independently developed client applications alongside research in medical AI and computer vision.

Core Technologies

Languages	Python, Java, Kotlin, JavaScript, TypeScript, SQL
Frontend	React, Next.js, Tailwind CSS, HTML, CSS
Backend	Node.js, Express.js, PostgreSQL, Prisma ORM, REST APIs
Mobile	Android SDK, Jetpack Compose, Room Database
Machine Learning / AI	PyTorch, OpenCV, Computer Vision, Deep Learning, timm, Grad-CAM
Cloud & DevOps	AWS (EC2, S3), Docker, Cloudflare, DigitalOcean, Railway, Git, Linux
Tools	GitHub, Android Studio, VS Code, Postman

Professional Experience

Software Engineer (Contract) Moonglade Communication

Aug 2023 – Present
Remote • India

Tech Stack: TypeScript • React • Next.js • Node.js • Express.js • PostgreSQL
AWS (EC2, S3) • Docker • Cloudflare • DigitalOcean

Domain: Marketing Technology (MarTech), Digital Campaigns, Consumer Engagement

- Designed, developed, and deployed production-grade digital campaign platforms for global consumer brands across Southeast Asia, supporting marketing campaigns and consumer engagement at scale.
- Built scalable full-stack applications, campaign management systems, and cloud infrastructure using modern web technologies, serving campaigns with over **20 million consumer interactions**.
- Successfully delivered multiple international software projects, leading to repeated contract engagements through consistent execution and reliable delivery.

AI Research Engineer Independent Research

Feb 2026 – Present
Remote • Germany

Tech Stack: Python • PyTorch • OpenCV • NumPy • Pandas • timm • Grad-CAM

- Developing deep learning techniques for automated diagnosis of pediatric hip disorders using medical imaging.
- Designed end-to-end machine learning pipelines including preprocessing, proxy-label generation, model training, and evaluation.
- Applied explainable AI techniques to improve model interpretability for clinical decision support.

Freelance Software Engineer Self-Employed

2022 – Present
Remote • India

Tech Stack: Java • Kotlin • Android • Jetpack Compose • React • Next.js
Node.js • Express.js • TypeScript • PostgreSQL • Docker

- Designed and delivered custom Android and full-stack web applications for clients across multiple domains.
- Developed scalable backend systems, REST APIs, authentication workflows, and relational database solutions tailored to client requirements.
- Managed the complete software development lifecycle, including architecture, implementation, deployment, and ongoing maintenance.

Projects

Beehive Tracker

Tech Stack: Node.js • Express.js • PostgreSQL • JavaScript • Geolocation API

- Developed a full-stack hive management platform featuring GPS-based geotagging, interactive location tracking, hazard warnings, community reporting, and scalable backend APIs with PostgreSQL for efficient hive and user management.

EDcryptFab

Tech Stack: Java • Swing • File I/O

- Built a desktop file encryption and decryption application with password-based security, drag-and-drop file support, and an intuitive workflow for secure file management.

AI-based DDH Detection

Tech Stack: Python • PyTorch • OpenCV • timm • Grad-CAM

- Developed an end-to-end deep learning pipeline for pediatric hip dysplasia detection from pelvic X-rays, incorporating automated proxy-label generation, medical image preprocessing, and explainable AI techniques including Grad-CAM for model interpretation.

Education

Bachelor of Science (Honours) in Physics
Babasaheb Bhimrao Ambedkar Bihar University

2019 – 2022
Muzaffarpur, Bihar

Higher Secondary (PCM)
Reet Lal Surdeep Yadav College

2017 – 2019
Muzaffarpur, Bihar